Bio-201: Photosynthesis & Cellular Respiration

Plant Physiology: Photosynthesis and Cellular Respiration Notes
Course: Bio-201 Advanced Plant Biology
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Section 1: Chloroplast Structure and Thylakoid Membrane

The chloroplast is the primary organelle responsible for oxygenic photosynthesis in eukaryotic autotrophs.
Enclosed by a double membrane consisting of outer and inner lipid bilayers, the interior contains a protein-rich fluid called stroma.
Suspended within the stroma is an intricate network of interconnected flattened disc sacs termed thylakoids.
Thylakoids are stacked into discrete columns called grana (singular: granum).
The thylakoid membrane segregates the stroma from an internal aqueous compartment termed the thylakoid lumen.
Embedded within the thylakoid membrane are light-harvesting pigment complexes, predominantly Chlorophyll a, Chlorophyll b, and accessory carotenoids.
Chlorophyll a exhibits peak absorption in the blue-violet (430 nm) and red (662 nm) wavelengths of the visible spectrum.

Section 2: The Light-Dependent Reactions

The light-dependent reactions occur specifically within the thylakoid membrane and convert radiant solar energy into chemical energy in the form of ATP and NADPH.
The sequence initiates when photon energy excites P680 reaction center molecules inside Photosystem II (PSII).
Excitation of P680 leads to photo-oxidation, releasing high-energy electrons that are transferred to the primary electron acceptor pheophytin.
To replenish electrons lost by P680, the Oxygen-Evolving Complex (OEC) catalyzes the photolysis of water molecules via the reaction:
2 H2O -> 4 H+ + 4 e- + O2.
Molecular oxygen (O2) is released as a photosynthetic byproduct into the atmosphere.
Electrons transit through plastoquinone (PQ), the cytochrome b6f complex, and plastocyanin (PC) before reaching Photosystem I (PSI).
Light excitation at PSI drives electron transfer through ferredoxin and Ferredoxin-NADP+ Reductase (FNR) to reduce NADP+ into NADPH in the stroma.

Section 3: ATP Synthase and Photophosphorylation

As electrons cascade through the cytochrome b6f complex, protons (H+) are actively translocated from the stroma into the thylakoid lumen.
Along with protons generated from water photolysis in the lumen, this creates a steep transmembrane proton electrochemical gradient (proton motive force, ΔpH).
Protons flow down their thermodynamic gradient from the lumen back to the stroma through the rotary channel of ATP Synthase (CF0-CF1 complex).
This catalytic phosphorylation converts Adenosine Diphosphate (ADP) and inorganic phosphate (Pi) into Adenosine Triphosphate (ATP).
This synthesis driven by light-generated proton gradients is termed photophosphorylation.

Section 4: The Calvin Cycle (Light-Independent Reactions)

The Calvin-Benson cycle takes place in the chloroplast stroma and synthesizes triose phosphate carbohydrates from atmospheric carbon dioxide (CO2).

The cycle comprises three enzymatic stages:

Stage 1: Carbon Fixation. The enzyme Ribulose-1,5-bisphosphate carboxylase/oxygenase (RuBisCO) catalyzes the carboxylation of ribulose-1,5-bisphosphate (RuBP), yielding unstable six-carbon intermediates that spontaneously cleave into two molecules of 3-phosphoglycerate (3-PGA).

Stage 2: Reduction. ATP and NADPH produced during light reactions phosphorylate and reduce 3-PGA into glyceraldehyde-3-phosphate (G3P).

Stage 3: RuBP Regeneration. For every six turns of the cycle, two G3P molecules exit to form glucose and complex polysaccharides, while remaining G3P molecules regenerate RuBP through ATP-dependent phosphorylation.

Section 5: Cellular Respiration Comparison

While photosynthesis is an endergonic anabolic pathway storing energy in carbon bonds, cellular respiration is an exergonic catabolic pathway.

Cellular respiration oxidizes hexose sugars to produce CO2, H2O, and 30 to 32 ATP molecules per glucose.

Glycolysis occurs in the cytosol, the Krebs cycle in the mitochondrial matrix, and oxidative phosphorylation along the inner mitochondrial cristae.

Whereas water is split and oxygen is released in photosynthesis, molecular oxygen serves as the terminal electron acceptor reduced to water in mitochondrial respiration.